

## CO3 - Assessment Tool 3: Resolution Algorithm Implementation

### Answers

#### Question 1 - Rain and Wet Ground

Symbols: R = It rains, W = Ground is wet.

Propositional logic:  $R \rightarrow W$ , R.

CNF:  $(\neg R \vee W)$ , (R).

Negated goal:  $\neg W$ .

Resolution: Resolve  $(\neg R \vee W)$  with  $(R) \rightarrow W$ . Resolve W with  $\neg W \rightarrow$  empty clause  $\square$ .

Conclusion: Goal proved — the ground is wet.

#### Question 2 - Student Assignment Submission

Symbols: S = Student submits assignment, M = Student receives internal marks.

Propositional logic:  $S \rightarrow M$ , S (Rahul).

CNF:  $(\neg S \vee M)$ , (S).

Negated goal:  $\neg M$ .

Resolution: Resolve  $(\neg S \vee M)$  with  $(S) \rightarrow M$ . Resolve M with  $\neg M \rightarrow$  empty clause  $\square$ .

Conclusion: Goal proved — Rahul receives internal marks.

#### Question 3 - Library Membership

Symbols: L = Person is a library member, B = Person can borrow books.

Propositional logic:  $L \rightarrow B$ , L (Priya).

CNF:  $(\neg L \vee B)$ , (L).

Negated goal:  $\neg B$ .

Resolution: Resolve  $(\neg L \vee B)$  with  $(L) \rightarrow B$ . Resolve B with  $\neg B \rightarrow$  empty clause  $\square$ . No further clauses can be generated.

Final conclusion: Proved — Priya can borrow books.

#### Question 4 - Placement Eligibility

Symbols: C = Student clears aptitude test, E = Student is eligible for placement.

Propositional logic:  $C \rightarrow E$ , C (Arun).

CNF clauses:  $(\neg C \vee E)$ , (C).

Negated goal:  $\neg E$ .

Resolution: Resolve  $(\neg C \vee E)$  with  $(C) \rightarrow E$ . Resolve E with  $\neg E \rightarrow$  empty clause  $\square$ .

Resolution tree:  $\square \leftarrow [E, \neg E]; E \leftarrow [(\neg C \vee E), C]$ .

Conclusion: Proved — Arun is eligible for placement.

#### Question 5 - Access Control System

Symbols: P = User enters correct password, A = User is authenticated, G = User is granted access.

Propositional logic:  $P \rightarrow A$ ,  $A \rightarrow G$ , P.

CNF:  $(\neg P \vee A)$ ,  $(\neg A \vee G)$ , (P).

Negated goal:  $\neg G$ .

Resolution steps: Resolve  $(\neg P \vee A)$  with (P)  $\rightarrow A$ . Resolve A with  $(\neg A \vee G) \rightarrow G$ . Resolve G with  $\neg G \rightarrow$  empty clause  $\square$ .

Explanation: password correctness gives authentication; authentication gives access; negating the goal and deriving  $\square$  confirms the goal cannot be false.

Final conclusion: Proved — the user is granted access.